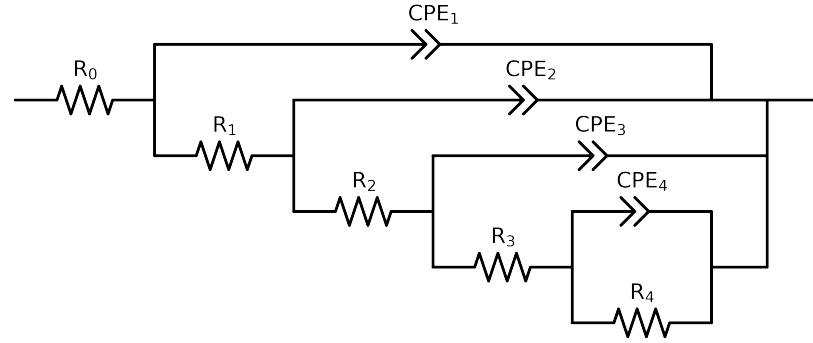


Comparative EIS Kinetics Report

Model Structure: R0-p(R1-p(R2-p(R3-p(R4,CPE4),CPE3),CPE2),CPE1)

Used data: original data



Element	Unit	Bounds	Vasco_LNV3_4	Vasco_LNV3_5	Vasco_LNV3_7d_
R_0	Ω	-	$1.56e+02 \pm 1.8e+00$	$1.25e+02 \pm 2.7e+00$	$1.20e+02 \pm 3.4e+00$
R_1	Ω	-	$4.67e+02 \pm 4.4e+01$	$8.28e+02 \pm 7.7e+01$	$9.87e+02 \pm 9.9e+01$
R_2	Ω	-	$2.73e+05 \pm 3.8e+04$	$4.19e+05 \pm 1.3e+05$	$3.64e+05 \pm 2.8e+05$
R_3	Ω	-	$3.65e+05 \pm 1.4e+05$	$6.01e+05 \pm 4.9e+05$	$7.14e+05 \pm 3.1e+05$
R_4	Ω	-	$4.14e+05 \pm 1.2e+05$	$3.29e+02 \pm 4.9e+05$	$1.84e+04 \pm 3.1e+05$
Q_4	$\Omega^{-1} \text{ sec}^a$	-	$3.84e-05 \pm 3.4e-05$	$1.33e-09 \pm 3.9e+03$	$1.10e-09 \pm 2.1e+03$
n_4	-	-	$9.90e-01 \pm 2.5e-01$	$8.63e-01 \pm 1.8e-04$	$1.00e+00 \pm 2.0e+06$
Q_3	$\Omega^{-1} \text{ sec}^a$	-	$1.87e-05 \pm 5.7e-06$	$4.54e-05 \pm 1.1e-03$	$5.03e-05 \pm 1.3e+00$
n_3	-	-	$1.00e+00 \pm 2.0e-01$	$1.00e+00 \pm 4.7e+00$	$1.00e+00 \pm 1.6e+00$
Q_2	$\Omega^{-1} \text{ sec}^a$	-	$3.70e-06 \pm 6.0e-07$	$5.96e-06 \pm 9.0e-07$	$7.50e-06 \pm 1.1e-06$
n_2	-	-	$9.11e-01 \pm 2.0e-02$	$8.75e-01 \pm 2.2e-02$	$8.54e-01 \pm 2.3e-02$
Q_1	$\Omega^{-1} \text{ sec}^a$	-	$3.61e-06 \pm 8.3e-07$	$5.85e-06 \pm 1.3e-06$	$6.74e-06 \pm 1.6e-06$
n_1	-	-	$7.72e-01 \pm 2.2e-02$	$6.87e-01 \pm 2.1e-02$	$6.59e-01 \pm 2.4e-02$
χ^2	-	-	$8.076e-04$	$1.513e-03$	$1.858e-03$

Analysis Results: Vasco_LNV3_4

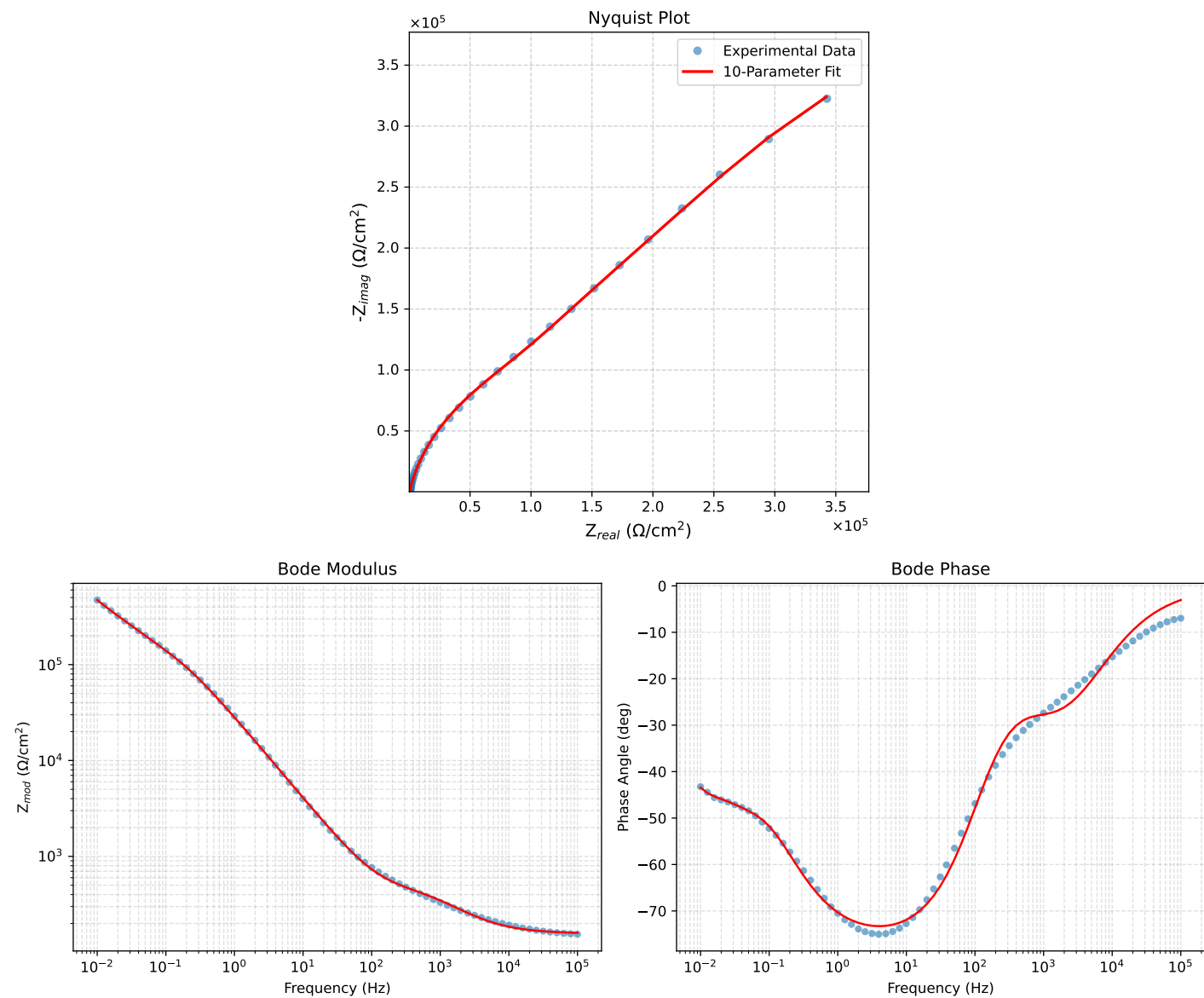


Figure 1: EIS Fit Results for Vasco_LNV3_4

Fitted Data: Vasco_LNV3_4

Frequency (Hz)	$\mathbf{z}_{real}(\Omega)$	$-\mathbf{z}_{imag}(\Omega)$	$\mathbf{z}_{mod}(\Omega)$	$\mathbf{z}_{phase}(\circ)$
1.0008e+05	1.5929e+02	8.4925e+00	1.5952e+02	-3.0518
7.9453e+04	1.5998e+02	1.0121e+01	1.6030e+02	-3.6201
6.3141e+04	1.6080e+02	1.2046e+01	1.6125e+02	-4.2842
5.0203e+04	1.6180e+02	1.4322e+01	1.6243e+02	-5.0583
3.9891e+04	1.6302e+02	1.7021e+01	1.6391e+02	-5.9605
3.1641e+04	1.6453e+02	2.0234e+01	1.6577e+02	-7.0111
2.5172e+04	1.6635e+02	2.3969e+01	1.6806e+02	-8.1993
2.0016e+04	1.6858e+02	2.8359e+01	1.7095e+02	-9.5489
1.5891e+04	1.7135e+02	3.3525e+01	1.7460e+02	-11.0702
1.2609e+04	1.7478e+02	3.9551e+01	1.7920e+02	-12.7508
1.0078e+04	1.7886e+02	4.6284e+01	1.8476e+02	-14.5081
8.0156e+03	1.8400e+02	5.4154e+01	1.9180e+02	-16.3998
6.3281e+03	1.9055e+02	6.3382e+01	2.0082e+02	-18.3984
5.0156e+03	1.9848e+02	7.3563e+01	2.1168e+02	-20.3359
3.9844e+03	2.0806e+02	8.4673e+01	2.2463e+02	-22.1447
3.1710e+03	2.1949e+02	9.6576e+01	2.3980e+02	-23.7498
2.5276e+03	2.3292e+02	1.0906e+02	2.5719e+02	-25.0895
1.9761e+03	2.4991e+02	1.2302e+02	2.7855e+02	-26.2087
1.5775e+03	2.6752e+02	1.3586e+02	3.0005e+02	-26.9243
1.2656e+03	2.8632e+02	1.4828e+02	3.2244e+02	-27.3792
9.9826e+02	3.0770e+02	1.6149e+02	3.4750e+02	-27.6915
7.9688e+02	3.2837e+02	1.7420e+02	3.7171e+02	-27.9455
6.2779e+02	3.4984e+02	1.8866e+02	3.9747e+02	-28.3363
5.0551e+02	3.6844e+02	2.0382e+02	4.2106e+02	-28.9511
3.9800e+02	3.8765e+02	2.2446e+02	4.4795e+02	-30.0724
3.1550e+02	4.0502e+02	2.5039e+02	4.7616e+02	-31.7252
2.5240e+02	4.2085e+02	2.8277e+02	5.0702e+02	-33.8974
1.9862e+02	4.3758e+02	3.2809e+02	5.4691e+02	-36.8621
1.5836e+02	4.5395e+02	3.8345e+02	5.9423e+02	-40.1880
1.2556e+02	4.7227e+02	4.5655e+02	6.5624e+02	-43.9741
1.0045e+02	4.9239e+02	5.4290e+02	7.3293e+02	-47.7931
7.9003e+01	5.1809e+02	6.6039e+02	8.3936e+02	-51.8850
6.3345e+01	5.4678e+02	7.9447e+02	9.6444e+02	-55.4630
5.0223e+01	5.8376e+02	9.6777e+02	1.1302e+03	-58.9015
3.8422e+01	6.3775e+02	1.2181e+03	1.3749e+03	-62.3649
3.1250e+01	6.8995e+02	1.4559e+03	1.6111e+03	-64.6443
2.4934e+01	7.6026e+02	1.7699e+03	1.9263e+03	-66.7545
1.9862e+01	8.4851e+02	2.1547e+03	2.3158e+03	-68.5060
1.5625e+01	9.6549e+02	2.6512e+03	2.8215e+03	-69.9895
1.2401e+01	1.1074e+03	3.2358e+03	3.4201e+03	-71.1071
9.9311e+00	1.2779e+03	3.9168e+03	4.1200e+03	-71.9299
7.9449e+00	1.4917e+03	4.7426e+03	4.9716e+03	-72.5394
6.3174e+00	1.7674e+03	5.7683e+03	6.0330e+03	-72.9649
5.0080e+00	2.1206e+03	7.0281e+03	7.3411e+03	-73.2101
3.9457e+00	2.5826e+03	8.5991e+03	8.9786e+03	-73.2830
3.1587e+00	3.1312e+03	1.0369e+04	1.0831e+04	-73.1966
2.5040e+00	3.8608e+03	1.2589e+04	1.3168e+04	-72.9507
1.9981e+00	4.7687e+03	1.5178e+04	1.5909e+04	-72.5577
1.5847e+00	5.9673e+03	1.8354e+04	1.9300e+04	-71.9893
1.2669e+00	7.4577e+03	2.1997e+04	2.3227e+04	-71.2718
9.9904e-01	9.5065e+03	2.6571e+04	2.8221e+04	-70.3141
7.9234e-01	1.2108e+04	3.1820e+04	3.4046e+04	-69.1678
6.3345e-01	1.5343e+04	3.7685e+04	4.0689e+04	-67.8470
5.0403e-01	1.9567e+04	4.4512e+04	4.8623e+04	-66.2705
4.0064e-01	2.4951e+04	5.2202e+04	5.7859e+04	-64.4540
3.1672e-01	3.1850e+04	6.0849e+04	6.8681e+04	-62.3713
2.5202e-01	4.0040e+04	6.9850e+04	8.0513e+04	-60.1772
2.0032e-01	4.9761e+04	7.9294e+04	9.3615e+04	-57.8897
1.5890e-01	6.0959e+04	8.9094e+04	1.0795e+05	-55.6197
1.2601e-01	7.3326e+04	9.9206e+04	1.2336e+05	-53.5309
1.0016e-01	8.6483e+04	1.0978e+05	1.3976e+05	-51.7703
7.9449e-02	1.0065e+05	1.2151e+05	1.5778e+05	-50.3645
6.3174e-02	1.1580e+05	1.3466e+05	1.7761e+05	-49.3064
5.0134e-02	1.3272e+05	1.4983e+05	2.0016e+05	-48.4660
3.9826e-02	1.5169e+05	1.6694e+05	2.2556e+05	-47.7400
3.1630e-02	1.7314e+05	1.8609e+05	2.5418e+05	-47.0657
2.5121e-02	1.9728e+05	2.0752e+05	2.8633e+05	-46.4496
1.9955e-02	2.2461e+05	2.3178e+05	3.2275e+05	-45.8996
1.5852e-02	2.5638e+05	2.5949e+05	3.6478e+05	-45.3448
1.2591e-02	2.9472e+05	2.9068e+05	4.1395e+05	-44.6042
1.0001e-02	3.4217e+05	3.2396e+05	4.7120e+05	-43.4339

Analysis Results: Vasco_LNV3_5

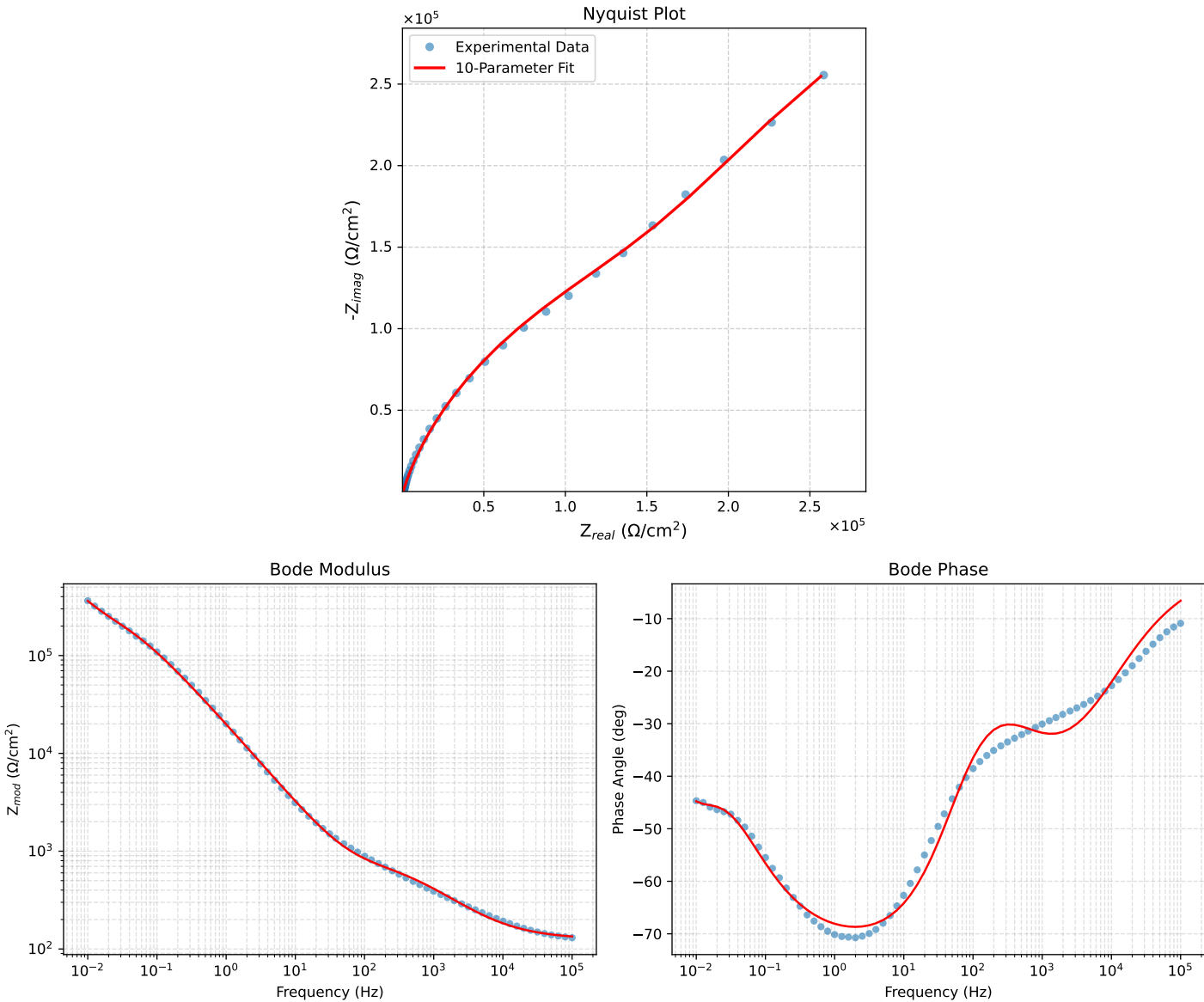


Figure 2: EIS Fit Results for Vasco_LNV3_5

Fitted Data: Vasco_LNV3_5

Frequency (Hz)	$\mathbf{z}_{real}$ (Ω)	$-\mathbf{z}_{imag}$ (Ω)	$\mathbf{z}_{mod}$ (Ω)	$\mathbf{z}_{phase}$ ($^{\circ}$)
1.0008e+05	1.3338e+02	1.5402e+01	1.3427e+02	-6.5871
7.9453e+04	1.3490e+02	1.7982e+01	1.3609e+02	-7.5928
6.3141e+04	1.3669e+02	2.0968e+01	1.3829e+02	-8.7210
5.0203e+04	1.3880e+02	2.4422e+01	1.4093e+02	-9.9794
3.9891e+04	1.4129e+02	2.8433e+01	1.4412e+02	-11.3778
3.1641e+04	1.4427e+02	3.3105e+01	1.4802e+02	-12.9236
2.5172e+04	1.4776e+02	3.8424e+01	1.5267e+02	-14.5765
2.0016e+04	1.5190e+02	4.4546e+01	1.5830e+02	-16.3443
1.5891e+04	1.5685e+02	5.1605e+01	1.6512e+02	-18.2121
1.2609e+04	1.6274e+02	5.9678e+01	1.7334e+02	-20.1384
1.0078e+04	1.6951e+02	6.8538e+01	1.8284e+02	-22.0152
8.0156e+03	1.7770e+02	7.8728e+01	1.9436e+02	-23.8950
6.3281e+03	1.8776e+02	9.0515e+01	2.0843e+02	-25.7382
5.0156e+03	1.9949e+02	1.0339e+02	2.2469e+02	-27.3962
3.9844e+03	2.1320e+02	1.1737e+02	2.4338e+02	-28.8339
3.1710e+03	2.2915e+02	1.3238e+02	2.6464e+02	-30.0159
2.5276e+03	2.4755e+02	1.4826e+02	2.8855e+02	-30.9178
1.9761e+03	2.7065e+02	1.6632e+02	3.1767e+02	-31.5714
1.5775e+03	2.9477e+02	1.8330e+02	3.4711e+02	-31.8745
1.2656e+03	3.2107e+02	1.9999e+02	3.7826e+02	-31.9179
9.9826e+02	3.5217e+02	2.1774e+02	4.1404e+02	-31.7276
7.9688e+02	3.8391e+02	2.3416e+02	4.4969e+02	-31.3808
6.2779e+02	4.1916e+02	2.5111e+02	4.8862e+02	-30.9251
5.0551e+02	4.5191e+02	2.6644e+02	5.2461e+02	-30.5230
3.9800e+02	4.8803e+02	2.8411e+02	5.6470e+02	-30.2061
3.1550e+02	5.2231e+02	3.0330e+02	6.0399e+02	-30.1438
2.5240e+02	5.5403e+02	3.2528e+02	6.4247e+02	-30.4180
1.9862e+02	5.8663e+02	3.5496e+02	6.8566e+02	-31.1775
1.5836e+02	6.1626e+02	3.9121e+02	7.2995e+02	-32.4079
1.2556e+02	6.4606e+02	4.3937e+02	7.8130e+02	-34.2184
1.0045e+02	6.7506e+02	4.9898e+02	8.3945e+02	-36.4703
7.9003e+01	7.0796e+02	5.8118e+02	9.1596e+02	-39.3831
6.3345e+01	7.4122e+02	6.7682e+02	1.0037e+03	-42.3998
5.0223e+01	7.8100e+02	8.0218e+02	1.1119e+03	-45.7668
3.8422e+01	8.3576e+02	9.8502e+02	1.2918e+03	-49.6865
3.1250e+01	8.8674e+02	1.1595e+03	1.4597e+03	-52.5932
2.4934e+01	9.5375e+02	1.3900e+03	1.6858e+03	-55.5445
1.9862e+01	1.0362e+03	1.6719e+03	1.9669e+03	-58.2092
1.5625e+01	1.1437e+03	2.0338e+03	2.3334e+03	-60.6485
1.2401e+01	1.2721e+03	2.4574e+03	2.7672e+03	-62.6313
9.9311e+00	1.4239e+03	2.9472e+03	3.2731e+03	-64.2133
7.9449e+00	1.6109e+03	3.5363e+03	3.8859e+03	-65.5091
6.3174e+00	1.8473e+03	4.2618e+03	4.6449e+03	-66.5651
5.0080e+00	2.1433e+03	5.1448e+03	5.5734e+03	-67.3835
3.9457e+00	2.5203e+03	6.2357e+03	6.7258e+03	-67.9927
3.1587e+00	2.9550e+03	7.4542e+03	8.0186e+03	-68.3757
2.5040e+00	3.5146e+03	8.9711e+03	9.6350e+03	-68.6061
1.9981e+00	4.1867e+03	1.0728e+04	1.1516e+04	-68.6803
1.5847e+00	5.0408e+03	1.2873e+04	1.3824e+04	-68.6151
1.2669e+00	6.0613e+03	1.5328e+04	1.6483e+04	-68.4236
9.9904e-01	7.4077e+03	1.8414e+04	1.9848e+04	-68.0860
7.9234e-01	9.0494e+03	2.1981e+04	2.3771e+04	-67.6232
6.3345e-01	1.1020e+04	2.6021e+04	2.8258e+04	-67.0478
5.0403e-01	1.3519e+04	3.0832e+04	3.3665e+04	-66.3238
4.0064e-01	1.6648e+04	3.6444e+04	4.0067e+04	-65.4489
3.1672e-01	2.0650e+04	4.3079e+04	4.7772e+04	-64.3892
2.5202e-01	2.5496e+04	5.0450e+04	5.6527e+04	-63.1892
2.0032e-01	3.1518e+04	5.8798e+04	6.6712e+04	-61.8071
1.5890e-01	3.8987e+04	6.8158e+04	7.8521e+04	-60.2299
1.2601e-01	4.8085e+04	7.8389e+04	9.1962e+04	-58.4741
1.0016e-01	5.8859e+04	8.9212e+04	1.0688e+05	-56.5845
7.9449e-02	7.1590e+04	1.0063e+05	1.2350e+05	-54.5707
6.3174e-02	8.5957e+04	1.1221e+05	1.4135e+05	-52.5465
5.0134e-02	1.0199e+05	1.2407e+05	1.6061e+05	-50.5786
3.9826e-02	1.1913e+05	1.3618e+05	1.8094e+05	-48.8219
3.1630e-02	1.3712e+05	1.4909e+05	2.0255e+05	-47.3950
2.5121e-02	1.5585e+05	1.6360e+05	2.2595e+05	-46.3895
1.9955e-02	1.7574e+05	1.8075e+05	2.5211e+05	-45.8053
1.5852e-02	1.9793e+05	2.0152e+05	2.8247e+05	-45.5155
1.2591e-02	2.2421e+05	2.2637e+05	3.1861e+05	-45.2748
1.0001e-02	2.5674e+05	2.5479e+05	3.6171e+05	-44.7813

Analysis Results: Vasco_LNV3_7d_

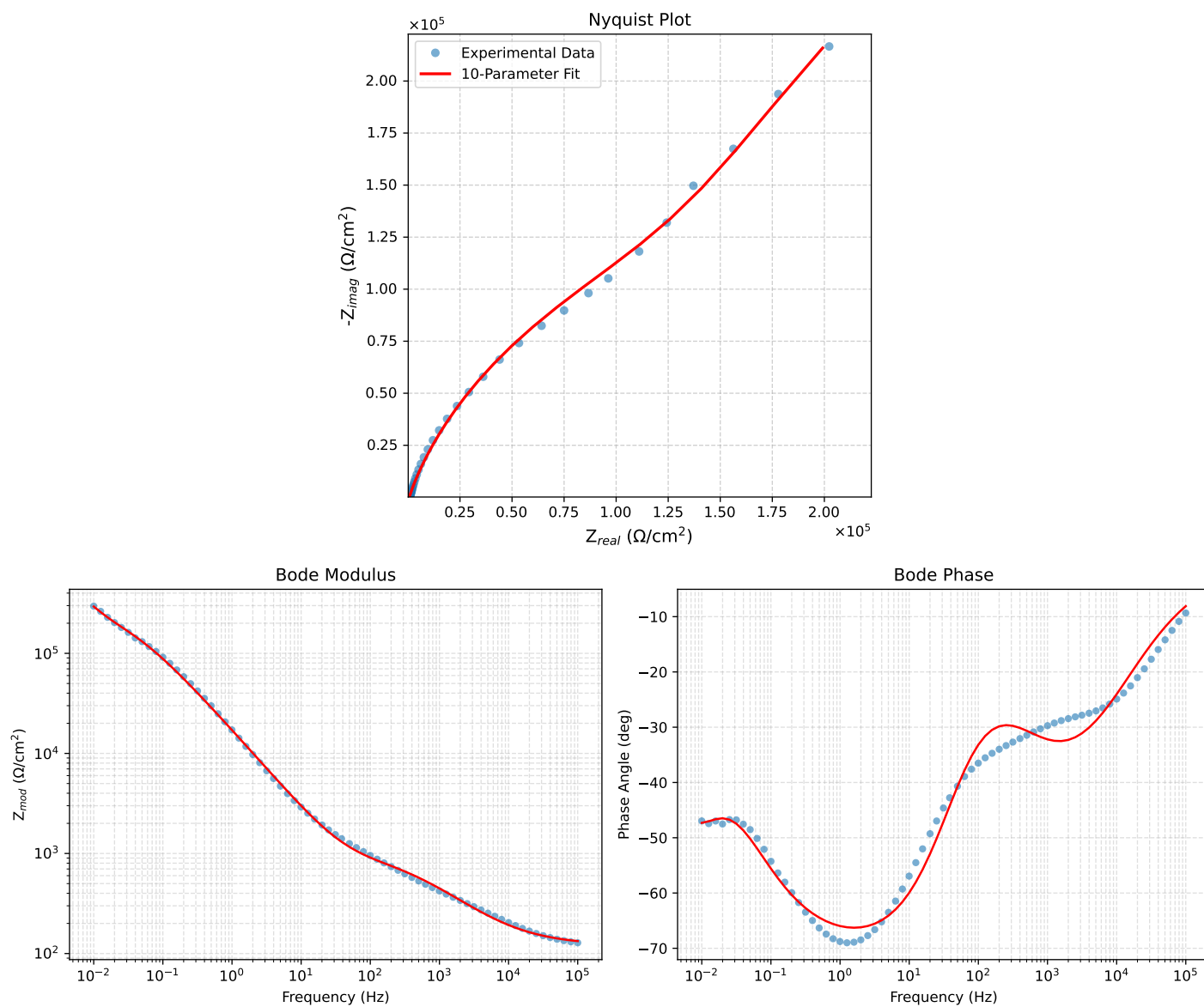


Figure 3: EIS Fit Results for Vasco_LNV3_7d_

Fitted Data: Vasco_LNV3_7d_

Frequency (Hz)	$\mathbf{z}_{real}$ (Ω)	$-\mathbf{z}_{imag}$ (Ω)	$\mathbf{z}_{mod}$ (Ω)	$\mathbf{z}_{phase}$ ($^\circ$)
1.0002e+05	1.3178e+02	1.8742e+01	1.3310e+02	-8.0946
7.9512e+04	1.3371e+02	2.1720e+01	1.3547e+02	-9.2263
6.3105e+04	1.3600e+02	2.5181e+01	1.3831e+02	-10.4900
5.0215e+04	1.3864e+02	2.9123e+01	1.4167e+02	-11.8630
3.9902e+04	1.4176e+02	3.3681e+01	1.4570e+02	-13.3656
3.1699e+04	1.4542e+02	3.8920e+01	1.5054e+02	-14.9835
2.5137e+04	1.4975e+02	4.4968e+01	1.5636e+02	-16.7141
1.9980e+04	1.5480e+02	5.1805e+01	1.6324e+02	-18.5036
1.5879e+04	1.6073e+02	5.9588e+01	1.7142e+02	-20.3416
1.2598e+04	1.6776e+02	6.8474e+01	1.8119e+02	-22.2041
1.0020e+04	1.7593e+02	7.8384e+01	1.9260e+02	-24.0151
7.9282e+03	1.8576e+02	8.9749e+01	2.0630e+02	-25.7876
6.3079e+03	1.9703e+02	1.0211e+02	2.2192e+02	-27.3947
5.0347e+03	2.1002e+02	1.1554e+02	2.3970e+02	-28.8169
3.9931e+03	2.2559e+02	1.3062e+02	2.6068e+02	-30.0719
3.1441e+03	2.4432e+02	1.4746e+02	2.8537e+02	-31.1122
2.5195e+03	2.6436e+02	1.6407e+02	3.1114e+02	-31.8249
2.0008e+03	2.8820e+02	1.8220e+02	3.4096e+02	-32.3008
1.5807e+03	3.1586e+02	2.0130e+02	3.7455e+02	-32.5098
1.2536e+03	3.4636e+02	2.2031e+02	4.1049e+02	-32.4597
1.0007e+03	3.7900e+02	2.3866e+02	4.4789e+02	-32.1993
7.9003e+02	4.1607e+02	2.5748e+02	4.8929e+02	-31.7513
6.2881e+02	4.5406e+02	2.7511e+02	5.3090e+02	-31.2112
5.0223e+02	4.9291e+02	2.9208e+02	5.7295e+02	-30.6494
4.0015e+02	5.3285e+02	3.0931e+02	6.1612e+02	-30.1342
3.1550e+02	5.7453e+02	3.2837e+02	6.6175e+02	-29.7502
2.5202e+02	6.1317e+02	3.4871e+02	7.0539e+02	-29.6269
2.0032e+02	6.5150e+02	3.7359e+02	7.5101e+02	-29.8315
1.5801e+02	6.8982e+02	4.0596e+02	8.0040e+02	-30.4769
1.2556e+02	7.2599e+02	4.4632e+02	8.5221e+02	-31.5821
9.9734e+01	7.6200e+02	4.9847e+02	9.1056e+02	-33.1913
7.9449e+01	7.9835e+02	5.6450e+02	9.7777e+02	-35.2634
6.2919e+01	8.3786e+02	6.5071e+02	1.0609e+03	-37.8338
4.9867e+01	8.8119e+02	7.5917e+02	1.1631e+03	-40.7457
3.8422e+01	9.3701e+02	9.1317e+02	1.3084e+03	-44.2619
3.1250e+01	9.8880e+02	1.0642e+03	1.4527e+03	-47.1035
2.4934e+01	1.0556e+03	1.2641e+03	1.6469e+03	-50.1375
2.0032e+01	1.1331e+03	1.4987e+03	1.8788e+03	-52.9080
1.5625e+01	1.2405e+03	1.8227e+03	2.2048e+03	-55.7603
1.2467e+01	1.3606e+03	2.1804e+03	2.5701e+03	-58.0354
9.9734e+00	1.5052e+03	2.6041e+03	3.0078e+03	-59.9713
7.9719e+00	1.6823e+03	3.1123e+03	3.5379e+03	-61.6071
6.3516e+00	1.9020e+03	3.7280e+03	4.1851e+03	-62.9697
5.0134e+00	2.1833e+03	4.4959e+03	4.9981e+03	-64.0977
3.9860e+00	2.5190e+03	5.3865e+03	5.9464e+03	-64.9365
3.1758e+00	2.9269e+03	6.4365e+03	7.0707e+03	-65.5470
2.5161e+00	3.4404e+03	7.7164e+03	8.4486e+03	-65.9698
1.9955e+00	4.0701e+03	9.2315e+03	1.0089e+04	-66.2074
1.5879e+00	4.8337e+03	1.1000e+04	1.2016e+04	-66.2790
1.2581e+00	5.7922e+03	1.3131e+04	1.4352e+04	-66.1979
9.9777e-01	6.9704e+03	1.5634e+04	1.7118e+04	-65.9707
7.9274e-01	8.4135e+03	1.8552e+04	2.0370e+04	-65.6049
6.2903e-01	1.0206e+04	2.1983e+04	2.4237e+04	-65.0964
4.9888e-01	1.2427e+04	2.5987e+04	2.8805e+04	-64.4426
3.9860e-01	1.5077e+04	3.0462e+04	3.3989e+04	-63.6675
3.1672e-01	1.8416e+04	3.5714e+04	4.0182e+04	-62.7219
2.5148e-01	2.2537e+04	4.1704e+04	4.7404e+04	-61.6127
1.9998e-01	2.7548e+04	4.8390e+04	5.5682e+04	-60.3473
1.5879e-01	3.3678e+04	5.5848e+04	6.5216e+04	-58.9090
1.2601e-01	4.1073e+04	6.4003e+04	7.6048e+04	-57.3101
1.0016e-01	4.9773e+04	7.2667e+04	8.8078e+04	-55.5909
7.9503e-02	5.9941e+04	8.1816e+04	1.0142e+05	-53.7725
6.3140e-02	7.1438e+04	9.1242e+04	1.1588e+05	-51.9406
5.0123e-02	8.4119e+04	1.0092e+05	1.3138e+05	-50.1888
3.9833e-02	9.7594e+04	1.1091e+05	1.4774e+05	-48.6543
3.1638e-02	1.1164e+05	1.2166e+05	1.6512e+05	-47.4609
2.5126e-02	1.2605e+05	1.3387e+05	1.8388e+05	-46.7223
1.9957e-02	1.4102e+05	1.4849e+05	2.0478e+05	-46.4777
1.5849e-02	1.5725e+05	1.6657e+05	2.2907e+05	-46.6480
1.2590e-02	1.7602e+05	1.8891e+05	2.5821e+05	-47.0239
1.0001e-02	1.9915e+05	2.1587e+05	2.9371e+05	-47.3067